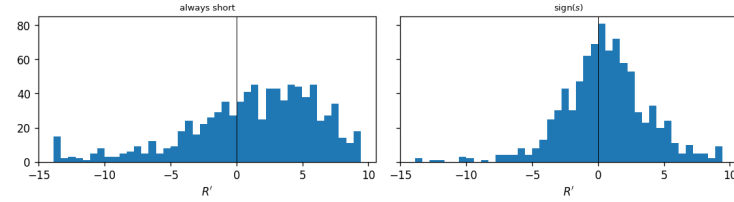


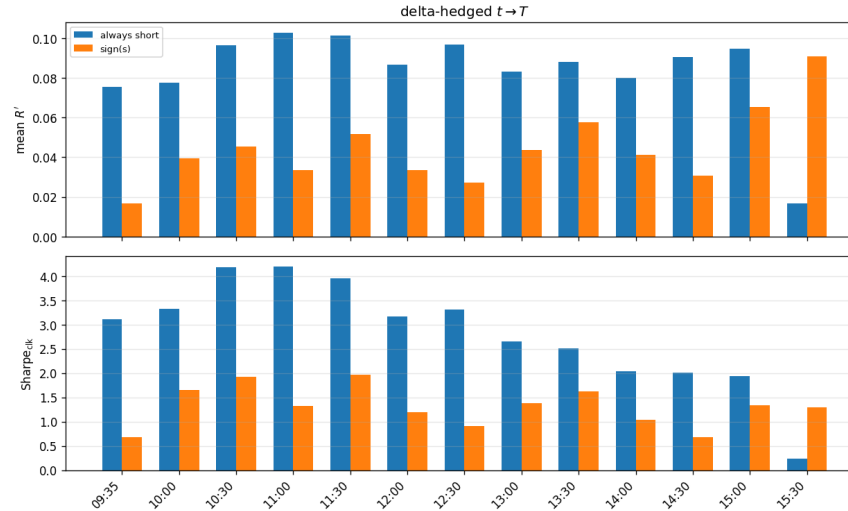
0DTE nearest-OTM straddle: per-rule return summary, pooled over the thirteen entry windows 09:35–15:30 ET (overlapping remaining-session books), models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	1.090	5.746	-58.68	10.56	-2.51	16.34	5.58	3.01	0	0.0	2.28
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.487	3.798	-17.72	52.74	2.69	42.06	3.78	2.04	295	34.1	0.93
block-diagonal ridge	866	0.576	3.874	-24.00	58.68	3.49	60.11	4.38	2.36	292	33.7	1.28
block-diagonal ridge, without the FOMC columns	866	0.548	3.828	-15.43	58.68	3.92	61.57	4.21	2.27	320	37.0	1.17
LightGBM [†]	866	0.573	3.888	-24.00	58.68	3.55	59.29	4.34	2.34	305	35.2	1.26
XGBoost [†]	866	0.564	3.904	-24.00	58.68	3.52	58.32	4.25	2.29	303	35.0	1.22
lasso (causally tuned) [†]	866	0.522	3.799	-24.00	52.74	2.61	43.15	4.04	2.18	320	37.0	1.07
lasso ($\alpha = 10^{-4}$)	866	0.579	3.635	-24.00	41.81	1.20	21.73	4.69	2.53	290	33.5	1.37
elastic net (causally tuned) [†]	866	0.493	3.900	-24.00	58.68	3.37	58.82	3.72	2.01	309	35.7	0.93

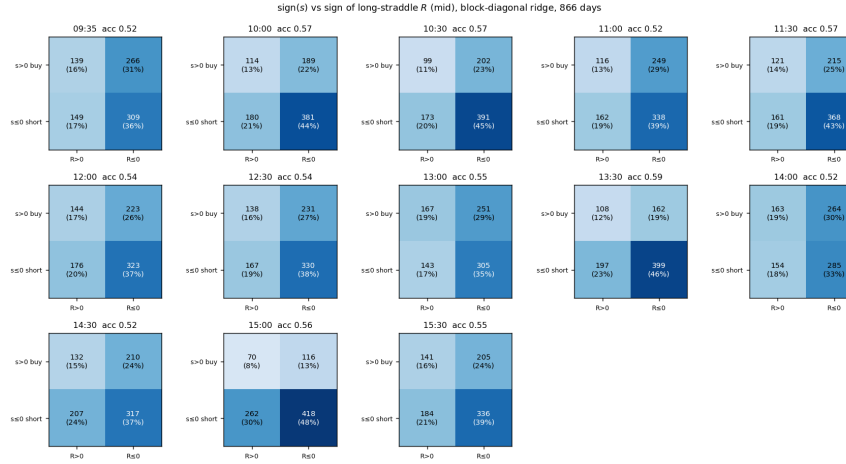
block-diagonal ridge, midpoint fills: daily sum over the thirteen entry windows 09:35-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the daily sum over the thirteen entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and Sharpe_{clk} by entry clock (grouped bars). Sharpe_{clk} is that clock's daily series $\times \sqrt{252}$, not the pooled Sharpe_{ann}. Delta-hedged $t \rightarrow T$: hold entry K to official close, rebalance Δ every 30 minutes (from 09:35 the first rebalance is at 10:00).



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum of the thirteen overlapping remaining-session books: enter at each clock 09:35–15:30 ET, hold those strikes to the official close, delta-hedge every 30 minutes (from 09:35 the first re-hedge is at 10:00). 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes**. Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s thirteen overlapping remaining-session books. That reserved name is the book’s daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^{\times}$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{ht}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar’s share, and h_t is the chain’s hours to the close.